Complex Analysis Homework 3

1. The *Jacobian* of a mapping $u = u(x, y)$, $v = v(x, y)$ from the xy-plane to the uv-plane is defined to be the determinant

$$J(x_0, y_0) := \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix},$$

where the partial derivatives are all evaluated at (x_0, y_0).

(a) Show that if $f = u + iv$ is analytic at $z_0 = x_0 + iy_0$, then $J(x_0, y_0) = |f'(z_0)|^2$.

(b) Let $f(z)$ be a bounded analytic function, defined on a bounded domain D in the complex plane, and suppose $f(z)$ is one-to-one. Show that the area of $f(D)$ is given by

$$\text{Area}(f(D)) = \iint_D |f'(z)|^2 \, dxdy.$$

(c) Sketch the image of the circle $\{|z-1| \le 1\}$ under the map $f(z) = z^2$. Compute the area of the image.

(d) Compute

$$\iint_D |f'(z)|^2 \, dxdy,$$

for $f(z) = z^2$ and D the open unit disk $\{|z| \le 1\}$. Interpret your answer in terms of areas.

Solution

(a)
$$\begin{aligned} J(x_0, y_0) &= \left. \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} \right|_{(x_0, y_0)} \\ &= \left[\frac{\partial u}{\partial x}(x_0, y_0)\right]^2 + \left[\frac{\partial v}{\partial x}(x_0, y_0)\right]^2 \\ &= |f'(z_0)|^2 \end{aligned}$$

(b) Recall

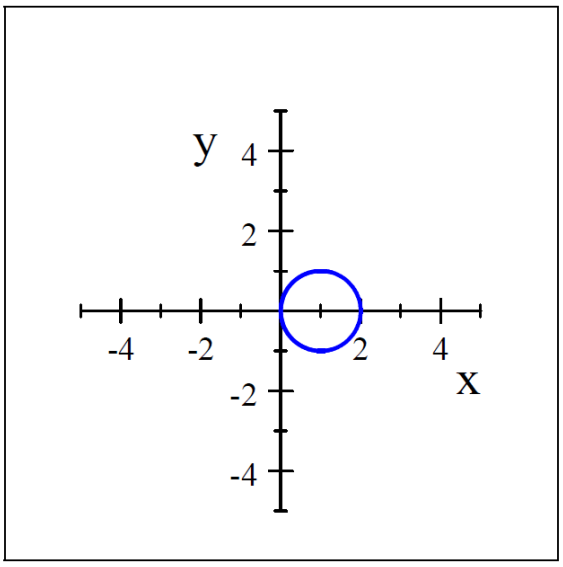
THEOREM 8 Change of Variables for Double Integrals

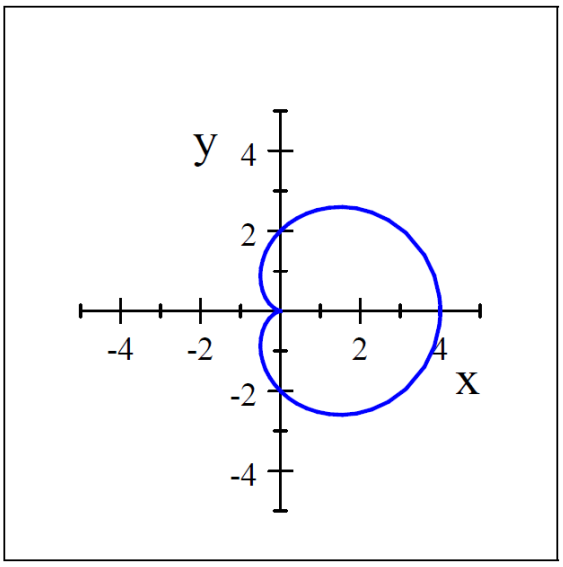
Let $T: x = g(u, v), y = h(u, v)$ be a transformation that maps a closed bounded region S in the uv-plane onto a region R in the xy-plane. Assume that T is one-to-one on the interior of S and that g and h have continuous first partial derivatives there. If f is continuous on R, then

$$\iint_R f(x, y)\, dA = \iint_S f(g(u, v), h(u, v))\, |J(u, v)|\, dA.$$

$$\text{Area}(f(D)) = \iint_{R=f(D)} 1 dxdy = \iint_D |J(x,y)|^2\, dxdy = \iint_D |f'(z)|^2\, dxdy.$$

(c)





Suppose $f(r_1 e^{i\theta_1}) = f(r_2 e^{i\theta_2}), \quad -\frac{\pi}{2} < \theta_1, \theta_2 < \frac{\pi}{2}$

$r_1^2 e^{2i\theta_1} = r_2^2 e^{2i\theta_2}$

$r_1^2 = r_2^2$ and $e^{2i\theta_1} = e^{2i\theta_2}$

$r_1 = r_2 \quad [\because r_1, r_2 > 0]$ and $2i\theta_1 - 2i\theta_2 = 2ki\pi$

$r_1 = r_2$ and $\theta_1 = \theta_2$

$f(z) = z^2$ is a 1-1 bounded analytic function from $\left\{z : -\frac{\pi}{2} < Arg(z) < \frac{\pi}{2}\right\}$ to $\{z : -\pi < Arg(z) < \pi\}$. The boundary of $\{|z-1| \le 1\}$ in polar coordinate is $r = 2\cos\theta$.

$$
\begin{aligned}
\text{Area}(f(D)) &= \iint_D |f'(z)|^2 \, dxdy \\
&= 4\iint_D |z|^2 \, dxdy \\
&= 4\int_{-\pi/2}^{\pi/2}\int_0^{2\cos\theta} r^3 dr d\theta \\
&= \int_{-\pi/2}^{\pi/2} \left[r^4 \right]_0^{2\cos\theta} d\theta \\
&= 16\int_{-\pi/2}^{\pi/2} \cos^4\theta d\theta \\
&= 16\left(\left[\frac{\cos^3\theta \sin\theta}{4} \right]_{-\pi/2}^{\pi/2} + \frac{3}{4}\int_{-\pi/2}^{\pi/2} \cos^2\theta d\theta \right) \\
&= 16\left(\frac{3}{4}\left[\frac{\cos\theta \sin\theta}{2} \right]_{-\pi/2}^{\pi/2} + \frac{3}{4}\cdot\frac{1}{2}\int_{-\pi/2}^{\pi/2} 1 d\theta \right) \\
&= 6\pi
\end{aligned}
$$

(d) $\iint_D |f'(z)|^2 \, dxdy = \iint_D 4|z|^2 \, dxdy = 4\int_0^{2\pi}\int_0^1 r^3 dr d\theta = \int_0^{2\pi} \left[r^4 \right]_0^1 d\theta = 2\pi$

The function f is a 2-1 mapping on $C\backslash\{0\}$. It maps the top and bottom halves of the unit disk one-to-one onto the unit disk, so by (b) each of the integrals on top and bottom halves are π.

2(a) Evaluate $(-1+i)^i$, i^i and $(-1-i)^i$.

(b) As we all know, $(-1+i)\,i = -1-i$. Is $\left(\text{principal value of } (-1+i)^i\right) \times \left(\text{principal value of } i^i\right)$ $= \left(\text{principal value of } (-1-i)^i\right)$ correct?

(c) Is there any equality in $(-1+i)^{2i}$, $\left((-1+i)^2\right)^i$, $(-1+i)^i \times (-1+i)^i$?

Solution

(a) $(-1+i)^i = e^{i\log(-1+i)} = e^{i\left(\text{Log}\sqrt{2} + i\pi(2k+3/4)\right)} = e^{(-3/4-2k)\pi + i\text{Log}\sqrt{2}}$

$i^i = e^{i\log i} = e^{i(i\pi(2k+1/2))} = e^{(-1/2-2k)\pi}$

$(-1-i)^i = e^{i\log(-1-i)} = e^{i\left(\text{Log}\sqrt{2} + i\pi(2k-3/4)\right)} = e^{(3/4+2k)\pi + i\text{Log}\sqrt{2}}$

(b) No. It is because $e^{(-3/4)\pi + i\text{Log}\sqrt{2}} e^{(-1/2)\pi} = e^{(-5/4)\pi + i\text{Log}\sqrt{2}} \neq e^{(3/4)\pi + i\text{Log}\sqrt{2}}$

(c) None of them are equal.

$$(-1+i)^{2i} = e^{2i\log(-1+i)} = e^{2i\left(\mathrm{Log}\sqrt{2}+i\pi(2k+3/4)\right)} = e^{(-3/2-4k)\pi+i\mathrm{Log}2}$$
$$\left((-1+i)^2\right)^i = \left(1-2i+i^2\right)^i = \left(-2i\right)^i = e^{i\log(-2i)} = e^{i(\mathrm{Log}2+i\pi(2k-1/2))} = e^{(1/2-2k)\pi+i\mathrm{Log}2}$$
$$(-1+i)^i \times (-1+i)^i = e^{(-3/4-2k)\pi+i\mathrm{Log}\sqrt{2}} \times e^{(-3/4-2l)\pi+i\mathrm{Log}\sqrt{2}} = e^{(-3/2-2m)\pi+i\mathrm{Log}2}$$

3. Solve $\sin z = \dfrac{1}{\sin 2\theta}$ where $0 < \theta < \frac{\pi}{2}$.

Solution

$$\sin z = \frac{1}{\sin 2\theta}$$
$$\frac{e^{iz} - e^{-iz}}{2i} = \frac{1}{\sin 2\theta}$$
$$e^{iz}\sin 2\theta - 2i - e^{-iz}\sin 2\theta = 0$$
$$e^{2iz}\sin 2\theta - 2ie^{iz} - \sin 2\theta = 0$$
$$e^{iz} = \frac{2i \pm \sqrt{(-2i)^2 + 4\sin^2 2\theta}}{2\sin 2\theta} = \frac{1 \pm \cos 2\theta}{\sin 2\theta} i = \frac{2\cos^2\theta}{2\sin\theta\cos\theta} i \quad \text{or} \quad \frac{2\sin^2\theta}{2\sin\theta\cos\theta} i$$
$$= i\cot\theta \quad \text{or} \quad i\tan\theta$$
$$iz = \pm\mathrm{Log}(\tan\theta) + \left(2k + \frac{1}{2}\right)i\pi$$
$$z = \left(2k + \frac{1}{2}\right)\pi \pm i\mathrm{Log}(\tan\theta)$$

4. Solve $z^4 - (5-14i)z^2 - 2(5i+12) = 0.$

Solution

Let $w = z^2$. We need to solve $w^2 - (5-14i)w - 2(5i+12) = 0.$

$(5-14i)^2+8(5i+12)=-75-100i=(u+iv)^2=(u^2-v^2)+2uvi.$

$\frac{u^2-v^2}{-75}=\frac{2uv}{-100}$

$2u^2-2v^2=3uv$

$2u^2-3uv-2v^2=0$

$(2u+v)(u-2v)=0$

$v=-2u$ or $0.5u$

$u+iv=5-10i.$ No solution for $v=0.5u$

$z^2=w=\frac{(5-14i)\pm(5-10i)}{2}=5-12i$ or $-2i$

$z^2=5-12i=(x^2-y^2)+2xyi$

$\frac{x^2-y^2}{5}=\frac{2xy}{-12}$

$-6(x^2-y^2)=5xy$

$6x^2+5xy-6y^2=0$

$(3x-2y)(2x+3y)=0$

$y=1.5x$ or $y=-\frac{2}{3}x$

$z=x+yi=\pm(3-2i)$

$z^2=-2i \Rightarrow z=\pm(1-i)$

5. Let $S_1=\{z\in\boldsymbol{C}:|z|<1\}$ and $S_2=\{z\in\boldsymbol{C}:\mathrm{Re}\,z<0\}$. Define a function $f:S_1\to S_2$ by $f(z)=\frac{z+i}{z-i}$.

(a) Show that Re($f(z)$) < 0 for all $z\in S_1$.

(b) Show that f is injective and surjective.

(c) Express the inverse function of f.

Solution

(a) $\mathrm{Re}\,f(z)=\frac{1}{2}\left(f(z)+\overline{f(z)}\right)=\frac{1}{2}\left(\frac{z+i}{z-i}+\frac{\bar{z}-i}{\overline{z-i}}\right)=\frac{1}{2}\left(\frac{(z+i)(\bar{z}+i)+(\bar{z}-i)(z-i)}{|z-i|^2}\right)=\frac{z\bar{z}-1}{|z-i|^2}<0$

(b) Suppose $f(z_1)=f(z_2)$. We have

$$\frac{z_1+i}{z_1-i}=\frac{z_2+i}{z_2-i}$$
$$(z_1+i)(z_2-i)=(z_2+i)(z_1-i)$$
$$iz_2-iz_1=iz_1-iz_2$$
$$2iz_1=2iz_2$$
$$z_1=z_2$$

For any $w\in S_2$, we need to find a point $z\in S_1$ such that $w=f(z)=\frac{z+i}{z-i}$. If this happens, then

$$w=\frac{z+i}{z-i} \iff w(z-i)=z+i \iff wz-iw=z+i \iff wz-z=iw+i \iff z=i\frac{w+1}{w-1}$$

Let us check.

$$f\left(i\frac{w+1}{w-1}\right)=\frac{i\frac{w+1}{w-1}+i}{i\frac{w+1}{w-1}-i}=\frac{w+1+(w-1)}{w+1-(w-1)}=w.$$

Furthermore,

$$|w+1|^2=w\bar{w}+w+\bar{w}+1=w\bar{w}+2\mathrm{Re}(w)+1<w\bar{w}-2\mathrm{Re}(w)+1=w\bar{w}-w-\bar{w}+1=|w-1|^2$$
$$|z|=\left|i\frac{w+1}{w-1}\right|=\left|\frac{w+1}{w-1}\right|<1$$
$$z\in S_1$$

(c) Define a function $g:S_2\to S_1$ by $g(w)=i\frac{w+1}{w-1}$. It is an inverse function of f. Clearly,

$f(g(w)) = w$.

$$g(f(z))=i\frac{f(z)+1}{f(z)-1}=i\frac{\frac{z+i}{z-i}+1}{\frac{z+i}{z-i}-1}=i\frac{z+i+z-i}{z+i-z+i}=z$$

g is an inverse function of f.